

CPU Performance Metrics and Analysis

1. Relative Performance

To compare how fast one computer is over another, we use the ratio of their execution times.

Example:

- Computer A takes 10 seconds
- Computer B takes 15 seconds

$$\text{Relative Performance} = \frac{\text{Execution Time}_B}{\text{Execution Time}_A} = \frac{15}{10} = 1.5$$

Conclusion: Computer A is **1.5 times faster** than Computer B.

2. Measuring Performance

Performance is often assessed by the time taken to complete a task.

Types of Time:

- **Elapsed Time:** Total time from start to end, including CPU, I/O, and waiting.
- **CPU Time:** Time CPU spends actively executing instructions.

3. Types of CPU Time

- **User CPU Time:** Time spent executing the program's instructions.
- **System CPU Time:** Time spent on behalf of the program by the OS (e.g., memory handling, I/O).

4. CPU Utilization Example

After running a program, you measure:

- User CPU Time = 90.7 s
- System CPU Time = 12.9 s
- Elapsed Time = 159 s

$$\text{CPU Utilization} = \frac{90.7 + 12.9}{159} = \frac{103.6}{159} \approx 0.652 = 65.2\%$$

Conclusion: The CPU was active **65.2%** of the time during execution.

5. Key Terms Recap

Term	Meaning
Elapsed Time	Total wall-clock time from start to finish of a program, includes CPU, I/O, and other delays.
CPU Time	Total time the CPU spent working on the task (User + System).
User CPU Time	Time spent running user-level code (your program).
System CPU Time	Time spent by the OS performing tasks for the program.
CPU Utilization	Percentage of elapsed time during which the CPU was actively working.